CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EE)

May 2014

EE-306 / EE 306.01 ELECTRICAL MACHINES III

Date: 6.5.2014, Tuesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the following question in context to synchronization of 3 phase alternator with infinite bus **(Any 4)** **[08]**

1.) Draw connection diagrams of lamps and corresponding diagram of resultant voltage across the lamps in 1 dark and 2 bright lamp method.

2.) How will you determine frequency difference between two alternators in 1 dark and two bright lamp methods?

3.) What are the advantages of using "Synchroscope" as compared to lamp methods?

4.) What will be the effect of increasing prime mover input of incoming alternator?

5.) Explain the concept of infinite bus and floating mode of operation of alternator.

6.) What are the conditions to be satisfied for parallel operation of alternator with infinite bus?

Q - 2 (a) Differentiate between hydro generator and turbo generators **[06]**

(b) Draw and explain synchronous generator capability curve. **[06]**

(c) Draw the equivalent circuit of an alternator on load. Also draw the vector diagram and write the emf equations for lagging leading and unity power factor loads. **[06]**

OR

Q - 2 (a) Draw and explain the in context to synchronous generators **[06]**

I. Load-Frequency charecteristic

II. P-δ curve

(b) Explain the advantages of rotating field winding and stationary armature winding in construction of synchronous generators. **[06]**

(c) Classify voltage regulation methods of 3 phase alternators and explain any one indirect method of voltage regulation in details. **[06]**

Q - 3 (a) Two three-phase alternators operate in parallel. The rating of one machine is 50 MW and that of the other is 100 MW. The droop characteristics of the governors of both machines are 4%. Assuming that the governors are operating at 50 Hz at no load, what will be the load shared by each machine if total load on system is 100 MW? **[04]**

(b) A 3 phase Y-connected alternator is rated at 1600 KVA, 13500 V. The armature effective resistance and synchronous reactance are 1.5 ohm and 30 ohm respectively per phase. Calculate the % regulation for a load of 1280 KW at p.f of **[05]**

i) 0.8 lagging. ii) Unity

OR

Q - 3 (a) 1.) Slip Test was conducted on salient pole SG. The maximum and minimum voltages observed are 2820 V and 2800 V respectively and maximum and minimum currents observed are 306 and 275 A, find out X_d and X_q of the machine. [02]

2.) Draw the vector diagram of Salient pole machine for lagging load connected to it. [03]

(b) A 3-phase, 16-Pole synchronous generator has a resultant air gap flux of 0.06Wb per pole. The flux is distributed sinusoidal over the pole. The stator has 2slots/pole/phase and 4 conductors /slot are accommodated in 2 layers. The coil span is 150 degree electrical. Calculate the phase and line induced voltages when the machine runs at 375 rpm. [04]

SECTION – II

Q – 4 What is SCR, of Synchronous machine? Derive the equation for the same and explain the effect of SCR on cost and operation of synchronous machine. [07]

Q - 5 (a) Explain the effect of varying the field excitation while keep the load on synchronous motor constant. Also draw the necessary vector diagram. [06]

(b) Draw and explain the 'V' curve, inverted 'V' curve, and "O" curve of the synchronous machine. [05]

(c) Write a note on cooling systems of synchronous machines. [03]

OR

Q - 5 (a) Explain the effect of varying the load of the synchronous motor and keeping the excitation constant, along with the necessary vector diagram. [06]

(b) Explain the phenomenon of hunting or phase swinging in synchronous motors. [04]

(c) Write down the difference between permanent magnet synchronous motor and linear synchronous motor. [04]

Q – 6 (a) State the advantages, disadvantages and applications of SRM motor. [06]

(b) Explain the concept, construction, working, advantages and applications of PMBLDC. [08]

OR

Q - 6 (a) Difference between PMDC, PMBLDC and Induction motor [05]

(b) Derive the voltage equation matrix for the three phase induction motor dynamic model [05]

(c) Explain Cross Field Theory of power amplification in DC Machines. [04]

****